1 High Level Summary

Randomized protocols are distributions over all protocols. There are two main sources of randomness, public and private randomness. Public randomness is when everyone shares a random string, and private randomness is where everyone has their own random string. Introducing randomness is an effective way at improving communication.

2 Public Randomness

2.1 Overview

Public randomness is when the randomness is shared amongst all parties, that is to say that there exists a random string that is shared with all parties.

2.2 Example

Say we have the most basic of problems, equality. Alice wants to see if she has the same input as Bob. Normally, this would take N+1 bits to determine if they are equal as Alice would send all of her bits to Bob who would then send a single bit to determine if they are equal. However, with shared randomness, it is possible to improve upon this by using a shared hashing function that maps N bits into K bits, which is exponentially better. Alice could then send her bits hashed to Bob who would then send a single bit back to Alice.

2.3 Applicability

This idea of shared randomness does seem to be very useful as we might be able to use shared functions to improve our bounds. However, it doesn't seem likely that the bounds would improve very much as our problem is not really limited by size of communication but rather the quantity of communication. There may be a way to implement it but it does not seem like any improvements would be made.

3 Private Randomness

3.1 Overview

Private randomness is when each party have their own unique random string.

3.2 Example

Just about every Public Randomness model can be replicated using private randomness, in this case, the party would just have to reveal what their random private string is. It is also important to note that private randomness is strictly weaker than public randomness. However, private randomness is better suited for real world applications as it shows a realistic constraint. Private randomness is still useful as even though we have to somehow communicate our randomness, it does not lose the main benefit of randomness in that it reduces the actual size of our information.

3.3 Applicability

It seems like this might be useful for our topic since we are looking at various parties who have limited information on each other. However, given that it is strictly weaker than the public protocol version, it may not be all that useful.

4 More Examples of how randomness can be used

4.1 Greater Than

Greater than has a deterministic communication complexity of log n bits, which is when Alice sends her number to Bob. However, using a randomized protocol, there is actually a protocol that requires only O(loglogn) bits. This can be done by using the same idea as equality, but checking to see if the n/2 most significant bits are equivalent, if so, you would do the same with the bottom n/2 bits. If they are different, in either half, then you would check the next significant half, which would be n/4 bits to see if they are equal. This process is continued until either there are no more bits to check the difference of, in which case it would be equal, or a difference is found.